Junmo Lee

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Google Scholar: https://scholar.google.com/citations?user=2xu0cTYAAAAJ&hl=en&oi=sra

SUMMARY

Device & process-integration engineer with proven end-to-end proficiency in next-generation advanced logic/memory technology development across academic and industry settings. Demonstrated track record spanning (i) device fabrication, (ii) process integration, (iii) device characterization, and (iv) device-to-system co-design for AI.

- Strong portfolio of pioneering fabrication and development of emerging non-volatile memory/DRAM devices, oxide TFTs, and their AI hardware applications - leveraging oxide semiconductor (W-doped indium oxide) and ferroelectric (HZO) materials.

- Led successful BEOL-compatible memory R&D projects in collaboration with TSMC, from ideation through experimental monolithic 3D integration of novel memories in the BEOL of foundry CMOS chips.

- Proficient in automated device and array characterization, including electrical and reliability analysis, using script-based measurement and data-analysis workflows with AI-assisted coding.

- Lead developer of NeuroSim V1.4, a device-to-system modeling framework for AI hardware at leading-edge technology nodes (14 nm–5 A), widely used in the device research community.

- Strong cross-functional device research capability demonstrated through multiple first-author publications at top device conferences (**4 IEDM 1 VLSI**), spanning device/array fabrication through device-to-system co-design for AI.

- Industry experience with advanced logic transistor and BEOL engineering, including compact modeling, within-wafer variation analysis at leading-edge nodes.

EDUCATION

Georgia Institute of Technology (Georgia Tech) — Atlanta, Georgia
Doctor of Philosophy — August 2022 – May 2027 (tentative)
Major in Electrical and Computer Engineering, Minor in Physics (Advisor: Dr. Shimeng Yu)
SRC research scholar
Cumulative GPA: 3.91/4.0

Seoul National University (SNU) — Seoul, South Korea
Bachelor of Science — March 2016 - August 2022
Major in Electrical and Computer Engineering
Cumulative GPA: 3.95/4.3 (overall) 3.95/4.3 (major), graduated summa cum laude

WORK/RESEARCH EXPERIENCE

Intel Corporation, Logic Technology Development (LTD) — Hillsboro, Oregon
Device Engineer/Data Scientist Intern — August 2026 – November 2026

- Physics-based compact modeling of BEOL parasitics at leading-edge nodes.
- Within-wafer variation modeling and characterization of FEOL advanced logic transistors.
- AI-assisted statistical analysis of device performance metrics.

Graduate Research Assistant @ Georgia Tech (Ph.D) — Atlanta, Georgia
Device Fabrication/Process Integration/Device-to-System Co-design for AI — August 2022 – May 2027 (tentative)

- Process development of ALD-grown W-doped In_2O_3 and Hafnia-based superlattice stacks.
- Reliability characterization of Hafnia-based superlattice capacitors, including TDDB, retention, and endurance.
- Fabrication and device-engineering of emerging non-volatile and cache memories: BEOL-compatible non-volatile ferroelectric capacitor (nvCAP) and oxide-channel capacitive gain-cell (Cap-GC) for next-generation AI hardware.
- Collaboration with TSMC on first experimental demonstration of monolithic 3D process integration flow for nvCAPs and Cap-GCs in the BEOL of TSMC 40-nm logic chips.
- Led the development of NeuroSim V1.4 software: device-to-system co-design framework for AI hardware at advanced technology nodes (14nm – 5A nodes), now widely adopted by the device research community.
- Device-to-system benchmarking of Cap-GC-based and nvCAP-based AI hardware and 3D DRAM, proving system-level performance gains over the existing memory technologies (SRAM, DRAM, RRAM).

Army Missile Command — Taean, South Korea
Republic of Korea Army — May 2018 – January 2020

- Earned the Excellence Award (ranked first) in Encryption Contest
- Discharged with army soldier award.

HONORS & AWARDS

(1) Award Nomination (2025): Roger A. Haken Best Student Paper Award at IEDM 25'
(2) Award Nomination (2024): Best Oral Presentation Award at DRC 24' (top 15 finalist paper)
(3) Highlight paper (2025): IEDM 25' work on "Monolithic 3D Integration of Dual-Gated ALD Oxide-Channel Non-Volatile Capacitive Memory on 40nm Si CMOS for Digital Compute-in-Memory" selected as conference highlight
(4) Media Coverage (2025): IEDM 25' work featured in Nature Electronics (https://www.nature.com/articles/s41928-025-01533-9) and Semiconductor Engineering (https://semiengineering.com/oxides-bring-low-leakage-transistors-to-leading-edge-memories/)
(5) Gold Prize in IT Creativity Challenge (2017): ranked 2nd in undergraduate student circuit design contest held by IEIE (The Institute of Electronics and Information Engineers)
(6) Excellent Tutor Award (2021): awarded to excellent peer tutors of the semester at Seoul National University

SKILLS

(1) Languages: English (professional working proficiency), Korean (native)
(2) Microfabrication: ALD (Zr-doped Hafnia, W-doped Indium Oxide), Photolithography, PVD (metal, oxide thin films), Dry etch (CCP/ICP-RIE), Mask design tools (Cadence Virtuoso, KLayout), Ellipsometry, RTP (rapid thermal processing/annealing), BEOL monolithic 3D process-integration of memory devices
(3) Device simulation: TCAD (Sentaurus, Silvaco)
(4) Device characterization: Proficient with performing & scripting electrical characterization using Keysight B1500A, Keithley 4200A, oscilloscopes, function generators, switch matrices, and semi-automatic probe stations (Summit 12K).
(5) Circuit simulation: Cadence Virtuoso ADE Explorer
(6) Programming/Software: Python, C++, MATLAB, JMP
(7) Code development: Lead developer of NeuroSim V1.4 - C++/Python-based chip-level benchmarking framework for emerging memory technologies at leading nodes (beyond 3 nm), now widely adopted by the device/circuit research community.

PUBLICATION SUMMARY

** **Refer to PUBLICATIONS** below for the referred papers ([1] [2].. etc.) associated with each project

(1) Device fabrication, characterization, and process integration of functional oxide thin-film capacitors/transistors
- Pioneered ferroelectric non-volatile capacitive memories with ALD oxide channel, and performed device engineering to optimize the trade-off between retention and capacitance memory window (IEDM 24' [4], IEDM 25' [1])
- Experimental demonstration of charge-domain in memory search using ferroelectric non-volatile capacitive array (IEDM 25' [3])
- First experimental demonstration of monolithic 3D BEOL integration of ferroelectric non-volatile capacitive memories on TSMC's 40 nm chip (IEDM 25' [1], TED 26' [5])
- Developed BEOL-compatible high-voltage superlattice capacitors and performed comprehensive reliability/electrical characterization, such as time dependent dielectric breakdown (TDDB) characterization to optimize reliability-capacitance density trade-offs (IEDM 23' [6], DRC 24' [12], TED 24' [10]).
- First experimental demonstration of BEOL-compatible DC-DC converters with oxide thin-film transistors and high-voltage superlattice capacitors (VLSI 24' [7])
- Fabricated a novel 3D DRAM device, the "Capacitive Gain Cell," and developed its bit-cost-scalable process-integration scheme and array-level operation (IEDM 26' [2])

(2) Device-to-system modeling and benchmarking of emerging device technologies
- Developed NeuroSim V1.4: System-level power/performance/area (PPA) benchmarking framework for emerging device and compute-in-memory technologies at advanced technology nodes (TCAS-I 24' [8], T-VLSI 25' [9], Nature Reviews EE 24' [11], DAC 24' [16])
- Performed DTCO of Digital Compute-in-Memory at leading-edge technology nodes down to 0.5 nm technology node (TCAS-I 24' [8], T-VLSI 25' [9])

(3) Device-to-system codesign of AI hardware technologies
- Designed a novel AI accelerator (Capacitive Digital-Compute-in-Memory) based on experimentally demonstrated non-volatile capacitive memories on TSMC's 40nm chip (IEDM 25' [1])
- Designed a novel 8T Digital Compute-in-Memory based on 0.5 nm CFET technology (T-VLSI 25' [9], APCCAS 24' [14])
- Designed a novel weight update algorithm for AI accelerators based on analog synaptic devices (JXCDC 22' [15])

PUBLICATIONS

*** indicates equal contribution**

[1] [Selected Publication] **J. Lee**, C. Zhang, S. Park, H. Park, T.-H. Kim, L. Liu, E. Ambrosi, M. Song, X. Bao, M.-F. Chang, S. Datta, S. Yu, "Monolithic 3D Integration of Dual-Gated ALD Oxide-Channel Non-Volatile Capacitive Memory on 40nm Si CMOS for Digital Compute-in-Memory," **IEEE International Electron Devices Meeting (IEDM) 2025**, San Francisco, USA. **(highlight paper, nominated for the best student paper award)**

[2] [Selected Publication] **J. Lee**, K. Lee, C. Zhang, T.-H. Kim, M. Tian, L. J.-C. Liu, M.-F. Chang, S. Datta, S. Yu, "Logic-Compatible Capacitive Gain Cell Monolithically Integrated on Si CMOS for Bit-Cost-Scalable 3D Digital Compute-in-Memory," **IEEE International Electron Devices Meeting (IEDM) 2026, accepted**.

[3] [Selected Publication] O. Phadke*, **J. Lee*,** P.-K. Hsu, C. Zhang, A. Khan, S. Datta, S. Yu, "Array-Level Demonstration of Charge-Domain In Memory Search using Back-End-of-Line Compatible Ferroelectric Nonvolatile Capacitors," **IEEE International Electron Devices Meeting (IEDM) 2025**, San Francisco, USA.

[4] [Selected Publication] **J. Lee**, C. Zhang, M. Shon, J. Read, S. Deng, O. Phadke, P. V. Ravindran, M. Tian, Y.-C. Luo, T.-H. Kim, A. I. Khan, S. Datta, S. Yu, "BEOL-compatible non-volatile capacitive synapse with ALD W-doped In2O3 semiconductor layer," **IEEE International Electron Devices Meeting (IEDM) 2024**, San Francisco, USA.

[5] [Selected Publication] **J. Lee**, C. Zhang, S. Park, H. Park, T.-H. Kim, L. Liu, E. Ambrosi, M. Song, X. Bao, M.-F. Chang, S. Datta, S. Yu, "Dual-Gated ALD Oxide-Channel Non-Volatile Capacitive Memory Monolithically Integrated on Foundry 40 nm CMOS for Capacitive Digital Compute-in-Memory," **IEEE Transactions on Electron Devices**, vol. 73, no. 5, pp. 3091-3101, 2026. **(invited paper as a top-ranked student paper at IEDM 2025)**

[6] [Selected Publication] S. Deng, J. Kwak, **J. Lee**, K. A. Aabrar, T.-H. Kim, S. Kirtania, G. Choe, C. Zhang, W. Li, O. Phadke, S. Yu, S. Datta, "BEOL compatible oxide power transistors for on-chip voltage conversion in heterogenous 3D (H3D) integrated circuits," **IEEE International Electron Devices Meeting (IEDM) 2023**, San Francisco, USA.

[7] [Selected Publication] S. Deng*, J. Kwak*, **J. Lee***, D. Chakraborty, J. Shin, O. Phadke, S. G. Kirtania, C. Zhang, K. A. Aabrar, S. Yu, S. Datta, "Demonstration of on-chip switched-capacitor DC-DC converters using BEOL compatible oxide power transistors and superlattice MIM capacitors," **IEEE Symposium on VLSI Technology and Circuits (VLSI) 2024**, Hawaii, USA. **(recognized as a top-ranked student paper at VLSI 2024)**

[8] [Selected Publication] **J. Lee**, A. Lu, W. Li, S. Yu, "NeuroSim V1.4: Extending technology support for digital compute-in-memory towards 1nm node," **IEEE Transactions on Circuits and Systems I: Regular Papers**, vol. 71, no. 4, pp. 1733-1744, 2024.

[9] [Selected Publication] **J. Lee**, M. Shon, F. Waqar, S. Yu, "3-D Digital Compute-in-Memory Benchmark With A5 CFET Technology: An Extension to Lookup-Table-Based Design," **IEEE Transactions on Very Large Scale Integration Systems, 2025**, accepted.

[10] [Selected Publication] S. Deng*, J. Kwak*, **J. Lee***, D. Chakraborty, J. Shin, O. Phadke, S. G. Kirtania, C. Zhang, K. A. Aabrar, S. Yu, S. Datta, "Monolithic switched-capacitor DC-DC voltage converters using BEOL-compatible oxide power transistors and superlattice MIM capacitors," **IEEE Transactions on Electron Devices**, vol. 71, no. 12, pp. 7999-8006, 2024. **(invited paper as a top-ranked student paper at VLSI 2024)**

[11] [Selected Publication] Anni Lu*, **J. Lee***, T.-H. Kim*, M. Karim, R. Park, H. Simka, S. Yu, "High-speed emerging memories for AI hardware accelerators," **Nature Reviews Electrical Engineering**, vol. 1, pp. 24-34, 2024.

[12] [Selected Publication] **J. Lee**, S. Deng, J. Kwak, M. Shon, S. Datta, S. Yu, "Optimization of backside of silicon-compatible high voltage superlattice capacitor for 12V-to-6V on-chip voltage conversion," **IEEE Device Research Conference (DRC) 2024**, College Park, MD. **(Nominated for the best oral presentation award)**

[13] S. Deng, H.-C. Chou, C. Zhang, J. Kwak, A. D. Asokan, E. Quezada, E. Sarkar, **J. Lee**, J. Shin, D. Chakraborty, V. Sathe, S. Yu, S. Datta, "High-efficiency BEOL-compatible on-chip switched capacitor DC-DC converter using planar and vertical oxide transistors with trench capacitors," **IEEE International Electron Devices Meeting (IEDM) 2025**, San Francisco, USA.

[14] M. Shon*, **J. Lee***, S. Yu, "3D digital compute-in-memory benchmark with A5 CFET technology," **IEEE Asia Pacific Conference on Circuits and Systems (APCCAS) 2024**, Taipei, Taiwan.

[15] **J. Lee**, J. Hwang, Y. Cho, M. K. Park, W. Y. Choi, S. Kim, J. H. Lee, "CRUS: A Hardware-Efficient Algorithm Mitigating Highly Nonlinear Weight Update in CIM Crossbar Arrays for Artificial Neural Networks," **IEEE Journal on Exploratory Solid-State Computational Devices and Circuits, 8(2)**, 145-154, 2022.

[16] A. Lu, **J. Lee**, Y.-C. Luo, H. Li, I. Young, S. Yu, "Digital CIM with noisy SRAM bit: A compact clustered annealer for large-scale combinatorial optimization," **ACM/IEEE Design Automation Conference (DAC) 2024**, San Francisco, CA.

[17] S. Deng, J. Kwak, **J. Lee**, S. Yu, S. Datta, "BEOL-compatible on-chip DC-DC converters," **IEEE International Interconnect Technology Conference (IITC) 2024**, San Jose, CA.

[18] G. Jeong, J. Kwak, M. Vadlamani, **J. Lee**, S. Yu, "Physics-Informed Neural Network-Assisted Compact Modeling for Process Variation in Non-Volatile Capacitive Crossbar Arrays," **IEEE International Reliability Physics Symposium (IRPS) 2026**, Tucson, AZ, USA, pp. 1–4.

[19] D. Yu, H. Hwang, B. Oh, **J. Lee**, S. Yu, T.-H. Kim, H. Kim, "A Unified Flash Memory Platform for Mode-Adaptive and Robust AI Computation," **Advanced Science**, e76619, 2026.

[20] J. Kwak, G. Choe, **J. Lee**, S. Yu, "Monolithic 3D transposable 3T embedded DRAM with back-end-of-line oxide channel transistor," **IEEE International Symposium on Circuits and Systems (ISCAS) 2024**, Singapore.

[21] J. Kwak, S. Deng, **J. Lee**, S. Datta, S. Yu, "Backside active power delivery with hybrid DC-DC converter enabled by amorphous oxide semiconductor transistors," **IEEE Transactions on VLSI Systems**, 2025.

[22] J. Kwak, S. Deng, C. Zhang, G. Jeong, O. Phadke, S.G. Kirtania, **J. Lee**, S. Datta, S. Yu, "Machine Learning-Assisted Compact Modeling of AC Stress-Induced Bias Temperature Instability in Top- and Bottom-Gate W-Doped In2O3 Channel Semiconductor Transistors," **IEEE Transactions on Electron Devices, 2025**.

[23] O. Phadke, P.-K. Hsu, Y.-C. Luo, J. Sharda, V. Garg, J. Read, **J. Lee**, H. Mulaosmanovic, S. Dunkel, S. Beyer, S. Yu, "Ternary content addressable memory with ferroelectric non-volatile capacitor," **IEEE Electron Device Letters 2025.**

[24] J. Kwak, C. Zhang, G. Jeong, O. Phadke, S. G. Kirtania, **J. Lee**, S. Datta, S. Yu, "Machine learning-assisted modeling of AC stress-induced bias temperature instability in oxide channel transistors for 2T gain cell eDRAM," **IEEE International Reliability Physics Symposium (IRPS) 2025**, Monterey, CA.

[25] M. Vadlamani, J. Jia, T. Xie, Y.-C. Luo, **J. Lee**, S. Li, S. Yu, "Capacitive Crossbar Array for Solving Matrix Equations in One-Shot," **IEEE Electron Device Letters**, vol. 46, no. 3, pp. 389-392, 2025.

[26] W. Li, **J. Lee**, S. Yu, "Optimization strategies for digital compute-in-memory from comparative analysis with systolic array," **IEEE International Conference on Artificial Intelligence Circuits & Systems (AICAS) 2023**, Hangzhou, China.

[27] W. Li, X. Zhang, **J. Lee**, F. L. Degertekin, S. Li, S. Yu, "Enabling ultra-low power ultrasound imaging with compute-in-memory sparse reconstruction accelerator," **IEEE Biomedical Circuits and Systems Conference (Bio-CAS) 2023**, Toronto, Canada.

[28] W. Shin, B. Choi, **J. Lee**, S. Hong, Y. Jeong, G. Jung, J. Park, D. Kim, C. Lee, J. H. Lee, "Sub-ppm Detection of NO2 gas using IGZO TFT-type Gas Sensor with Optimal Signal-to-Noise Ratio", **International Conference on Solid State Devices and Materials (SSDM) 2021**, virtual.

Under Review

[29] **J. Lee**, C. Zhang, S. Datta, S. Yu, "Pulse-Mode Operation and Reliability of BEOL-Compatible Ferroelectric Non-Volatile Capacitive Memories with Amorphous Oxide Semiconductor Channels," **IEEE Transactions on Nanotechnology** **(journal extension paper of IEDM 2024)**

TEACHING EXPERIENCE

Undergraduate Tutor (@ Seoul National University) Seoul, South Korea
Peer Tutor (Semiconductor Devices) September 2021 – December 2021

- Provided 30 hours of online lecture, received Excellent Tutor Award

Undergraduate Tutor (@ Seoul National University) Seoul, South Korea
Peer Tutor (Introduction to Electronic Circuits and Laboratory) September 2022 – December 2022

- Provided 24 hours of online lecture

SCHOLARSHIPS

(1) Merit-based Scholarship, Seoul National University (2020): provided to undergraduate students with excellent academic record
(2) Scholarship, Korea Electric Power Corporation (2021): provided to undergraduate students with excellent academic record